Draft guidance on transition from GTS to WIS2

World Meteorological Organization

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Standing Committee on Information Management and Technology (SC-IMT)[1]

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Chapter 1. Preamble

1.1. Purpose

The purpose of this document is to provide technical guidance and additional information for members to transition from GTS to WIS2, in implementing the practices, procedures, and specifications defined in the Manual on WIS (WMO No. 1060), Vol II.

1.2. Audience

This draft guidance document is provided primarily for GTS data producers, data consumers, and service providers.

1.3. Status of this document

This draft guidance is as an informal document. The content will be updated, modified and improved throughout the WIS2 transition phase as detailed specifications and procedures are validated through testing. The information provided herein will be consolidated into a new version of the Guide to WIS (WMO No. 1061) upon completion of the pilot phase. (TODO: validate)

[1] https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt

[2] https://community.wmo.int/governance/commission-membership/infcom

Chapter 2. Introduction

The WMO Information System 2.0 (WIS2) will be implemented according to the schedule provided in Figure 1. A one-year pilot phase will start at the end of 2022, with several countries collaborating in building the WIS2 infrastructure. Each Country will have a different role in the WIS2 framework and will implement a specific component.

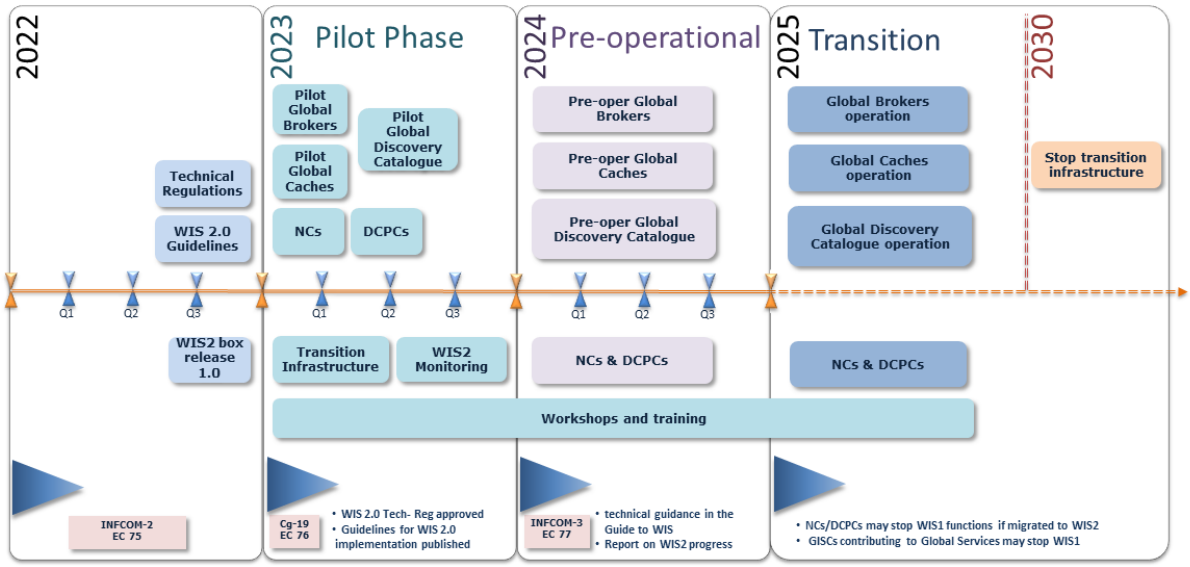

Figure 1. WIS2 implementation timeline

According to the WIS2 implementation plan (see figure 1), the Global Telecommunication System (GTS) will be decommissioned by 2030, and National Meteorological and Hydrological Services NMHSs will use the WIS2 platform for data exchange. During the transition period, there will be a mix of centres operating WIS2 and WIS/GTS. Some of them will be running both data-sharing frameworks simultaneously, with the difficulties associated with maintaining two operational systems for the same purpose. Therefore, a transition plan will be developed during the pilot phase, taking into account the time necessary for Members to migrate to the new systems and reducing, at the minimum, the time of parallel operation of both systems for a Member.

Chapter 3. Principles

The following principles are appropriate for the transition:

Principle 1: Each NMHS will be able to make the migration during the agreed period 2025-2030:

- Considering the fact that not all centres have the same skills and facilities, and the operational aspects of WIS and the risks involved in a big-bang approach, the WIS node implementation would not be fully implemented at the same time

Principle 2: No GTS data loss during the transition:

- During the pilot phase, and in coordination with the regional associations and GISCs, WIS2 infrastructure will be established in order to avoid data loss during the transition. The aim of this infrastructure is to ensure that data sent on the GTS can be received by a site having migrated on WIS2, and data, previously sent on the GTS, sent on WIS2 can be received by a site still on the GTS

Principle 3: Each Centre will decide when decommissioning the WIS1 and GTS:

- Decommissioning WIS1 services, as well as GTS ones, will be the decision of each NC/DCPC/GISC when they will consider that their and their user's migration is completed
- After migration to WIS2, It is not required to run an MSS to receive or send data from Centres not having made the transition. The Centre will decide when and if they want to stop their Message Switching System MSS. They can also stop the data dissemination to GTS

Principle 4: New data (eg. GBON, Climate, Hydrology, Cryosphere) will be exchanged solely on WIS2:

- WIS2 is designed to to enable the WMO Unified Data Policy, and to support the WMO Global Basic Observing Network. The new data will be available on WIS2. A centre not having made the migration to WIS2 will not receive the new data. This data will not have a GTS headers TTAAii and will not be exchanged over the GTS

Chapter 4. Pre-operational and operational phases

4.1. Definition of pre-operational phase

The objective of the pre-operational phase is to make the transition from the Pilot Phase to the operational phase.

The purpose of this phase is to verify that the WIS 2.0 implementation has been carried out in accordance with the technical specifications of WIS 2.0 and to make the changes approved in the pilot phase in a satisfactory operational manner.

This phase covers all activities related to the installation and integration of the WIS 2.0 components developed during the pilot phase, including the testing activities necessary to verify the integration and proper functioning of data exchange via WIS 2.0.

4.2. How to transition from the pilot to pre-operational phase

The pilot phase aims to develop the infrastructure of WIS 2.0 and make the tests to refine the WIS 2.0 technical specifications. The pre-operational phase is the clean-up phase to make the transition to operations.

The following actions are required for a smooth transition:

- software adopted to run a WIS 2.0 Node shall be compliant with WIS 2.0 technical regulations
- WIS 2.0 node details shall be communicated to the WMO Secretariat
- the WIS 2.0 Topic Hierarchy shall be updated to the latest approved version
- All data transmitted in GTS shall be published in WIS 2.0
- Discovery Metadata shall be prepared and published in accordance with the WMO Core Metadata Profile (WCMP2)
- SYNOP core data shall be published hourly (minutely data not required)

4.3. How to make the transition from the pre-operational to operational phase

The pre-operational phase aims to ensure that all the improvements made during the pilot phase are deployed and tested. During this phase performance tests of all the WIS 2.0 components will be made and relevant adjustments will be completed to be ready for the operation.

Members shall liaise with their users and make the necessary arrangements to meet their needs and point them to the WIS 2.0 global services.

The following actions shall be completed for a smooth transition from the pre-operational to operational phase:

- The latest technical improvements shall be installed and tested before **31 October 2024**.
- Build the skills needed to support the WIS 2.0 operations, which include:
  - Infrastructure
  - Upgrade management
  - Data management
  - Customer services support
  - Incident management
  - Monitoring
- Business continuity and disaster recovery: plans must be in place to ensure that, in the event of failure, the business can continue or, in the worst case, recover within the required time scale from any significant disaster. This might include:
  - provision for offsite backup
  - Supply of an identical replacement system within a specified time

Chapter 5. GTS to WIS2 Gateway

The Executive Council, with resolution 34 (EC-76), endorsed the WMO Information System 2 (WIS2) implementation plan and recognized the importance of establishing a pilot phase to develop the WIS2 infrastructure and begin testing it in order to be ready for a preoperational phase in 2024, and then for the transition starting on 2025. According to the WIS2 implementation plan, the Global Telecommunication System (GTS) will be decommissioned by 2030, and National Meteorological and Hydrological Services NMHSs will use the WIS2 platform for data exchange.

During the transition period, in order to avoid some WIS centres being forced to run both data-sharing frameworks WIS2 and GTS simultaneously, with the challenges associated with maintaining two operational systems for the same purpose, a gateway from GTS to WIS2 is designed, taking into account the time required for Members to migrate to the new systems and minimizing the time a member has to operate both systems in parallel.

5.1. Purpose

The purpose of the GTS to WIS2 gateway is to enable members who have migrated to WIS2 and have stopped their GTS systems to continue receiving GTS data from WIS2. This gateway also enables users who are not connected to GTS to access GTS data, during the transition phase, from WIS2. The GTS to WIS2 gateway will forward all the GTS traffic it receives to WIS2, that is each bulletin received on one of its GTS channels will be stored to the HTTP(s) endpoint of the Gateway and a WIS2 Notification Message is generated which points to that bulletin file. The Notification Message is published at the GTS to WIS2 gateway topic.

5.2. GTS to WIS2 Gateway provider

The gateway will be provided by Germany's Deutscher Wetterdienst (DWD) and the Japan Meteorological Agency (JMA).

5.3. Technical requirements

- GTS headers SHALL be used in the WIS2 Topic Hierarchy (WTH)
- The TH for GTS data on WIS2 SHALL be as follows:
  - `origin/a/wis2/[GC-name]/data/[core|recommended]/T1/T2/A1/A2/ii/CCCC`
  - e.g for DWD: `origin/a/wis2/de-dwd-gts-to-wis2/data/[core|recommended]/T1/T2/A1/A2/ii/CCCC` and for JMA: `origin/a/wis2/jp-jma-gts-to-wis2/data/[core|recommended]/T1/T2/A1/A2/ii/CCCC`
- The GTS to WIS2 data is copied on the WIS2 Global Caches if the data is WMO Core Data.
- Each GTS to WIS2 Gateway SHALL maintain a list of TTAAii headers for Core and for Recommended data in order to be able to send the notification to the correct topic. The list is coordinated and shared between the Gateways.
- The data_id for each GTS to WIS2 Gateway SHALL be unique. That is each gateway will choose different data ids for example by adding their centre name reflecting the fact, that GTS bulletins

with the same header might be different at the different locations. Therefore there will be two or more different GTS to WIS2 datasets (one for each gateway).

- Example for a valid GTS to WIS2 data id is `wis2/de-dwd-gts-to-wis2/data/core/W/W/X/X/60/EDZW/WWXX60EDZW200900`
- Data Consumers receiving the GTS data through WIS2 will need to be able to handle duplicates. This is in line with the current practices of handling duplicate messages on the GTS.
- Access to recommended GTS data should be restricted to WMO members.

Chapter 6. WIS2 to GTS Gateway

The WIS2 Implementation plan outlines a gradual transition of data exchange from the Global Telecommunications System (GTS) to WIS2. The transition is expected to occur between 2025 and 2030. The GTS will be decommissioned once the transition is complete.

Once a WIS centre has migrated from GTS to WIS2, subject to validation of readiness, the centre will cease to share data on the GTS. Consequently, during the transition period some data will be published on the GTS and some via WIS2.

WIS centres will need data published via both systems.

To remove the need for WIS centres to operate both GTS and WIS2 systems in parallel, real-time data published via WIS2 will be automatically re-published onto the GTS. The system responsible is referred to as the **WIS2 to GTS Gateway**.

To ensure resilient operation there will be more than one WIS2 to GTS Gateway.

6.1. Purpose

The purpose of the WIS2 to GTS Gateway is to enable Members relying on the GTS, who are yet to migrate to WIS2, to receive operational data published via WIS2 that would otherwise be unavailable on the GTS.

Note: Only data that have been already published in GTS are concerned by this Gateway, New data will be available **only** on WIS2.

6.2. WIS2 to GTS Gateway operators

The gateway will be provided by designated Regional Telecommunication Hubs (RTH). These RTHs are: [TODO: Add the RTHs that will each operate a WIS2-to-GTS Gateway. Provisional list is: China, UK, Brazil.]

6.3. Technical requirements

6.3.1. For Data Providers

A Member seeking to stop GTS transmission shall contact the WMO Secretariat, providing the following information:

1. The proposed date that GTS transmission will stop.
2. The WIS2 centre-id from which data destined for GTS transmission will be published.
3. The GTS header CCCC code to be used for the bulletins published on their behalf.
4. The list of GTS bulletins, and, where appropriate, their cut-off times, to be published by the Gateway on their behalf.

Once the Secretariat is satisfied that the Member’s WIS2 Node is operationally ready, they will

confirm the date on which the Member will stop published to the GTS. On this date the WIS2 to GTS Gateways will take over.

When publishing a WIS2 Notification Message for data that should also be shared on the GTS, the Data Provider shall indicate the GTS Abbreviated Header (AHL) of the bulletin in which the data is to be published.

This is done by including the `gts` property in the WIS2 Notification Message (see example below). This enables the WIS2 to GTS Gateway operator to easily identify data for republication on the GTS and the AHL of the associated bulletin.

```
"properties": {
  ...
  "gts": {
       "ttaaii": "FTAE31",
       "cccc": "VTBB"
  }
}
```

6.3.2. For WIS2 to GTS Gateway operators

A WIS2 to GTS Gateway operator shall operate the following components throughout the transition period:

1. A WIS2 client application to retrieve data published on WIS2.
2. A Message Switching System (MSS), with routing configuration to all other RTHs on the MTN.
3. Connectivity with the GTS Main Trunk Network (MTN).

Each WIS2 to GTS Gateway shall republish data on behalf of all Members who have stopped publishing to the GTS.

When a Member requests to stop their GTS transmission, the Secretariat will provide the following information to each WIS2 to GTS Gateway operator:

1. The `centre-id` of the WIS2 Node from which the data destined for GTS transmission will be published on WIS2.
2. The GTS header **CCCC** code to be used when publishing bulletins on behalf of that WIS2 Node.
3. The list GTS bulletins and cut-off times the Member requires to be published.

The Secretariat will set a mutually acceptable date for transfer of responsibility from Member to Gateways for GTS publication, allowing adequate set-up time for the Gateway operators. On that date, the Gateway operators will take over responsibility for publication of data on the GTS for the Member.

For each Member whose GTS transmission responsibility is passed to the WIS2 to GTS Gateway, the Gateway will:

1. Subscribe to notifications on the topic: `cache/a/wis2/{centre-id}/data/core/#`, where `{centre-`

`id}` refers to the WIS2 Node designated by the Member

2. Add the **CCCC** code designated by the Member to the Gateway's publication *whitelist*.
3. Complete configuration required in the MSS for each bulletin specified by the Member; e.g., set up the data import channel, set the cut-off time, etc.

Note: To ensure resilient operation, the Gateway should subscribe to notification messages from multiple Global Brokers.

Note: During the transition period, other gateways will republish GTS data to WIS2. These *GTS to WIS2 Gateways* will publish via a designated `centre-id`. To avoid an infinite loop of republication, it is essential that a WIS2 to GTS Gateway **does not** subscribe to notification messages associated with a centre-id of a GTS to WIS2 Gateway.

For each WIS2 notification message received, the Gateway will:

1. Parse the notification message, discarding it if:
   a. The notification message is a duplicate.
   b. The notification message refers to data that has already been republished by the Gateway.
   c. The notification message does not include the `gts` property.
   d. The value of the `cccc` property is not on the Gateway's publication whitelist.
2. If the notification message was not discarded, then:
   a. Parse the notification message to extract the **TTAAii** and **CCCC** which identifies the bulletin that the data should be published within.
   b. Download the data indicated in the message, using the *canonical* link if one is specified.
   c. Pass the downloaded data file to the Gateway's MSS via the channel configured for the bulletin (**TTAAii**, **CCCC**).

Note: the WIS2 to GTS Gateway should log and report failures. [TODO: elaborate on failure reporting]

The Gateway's MSS will process incoming data files, batching individual items into bulletins as per configuration, and publish those bulletins onto the GTS for onward distributed via RTHs on the MTN and beyond.

The Gateway's MSS will publish bulletin amendments / corrections where data arrives after the cut-off time.

Chapter 7. Management of WIS1 and GTS

7.1. Maintenance and operation of Message Switching System (MSS)

7.1.1. Main Telecommunication Network

With the start of the pre-operational phase of WIS 2.0, the Main Telecommunication Network (MTN), linking together the World Meteorological Centres (WMCs) as well as designated Regional Telecommunication Hubs (RTHs); shall keep their Message Switching Systems (MSS) up and continue to publish data collecting the bulletins from their associated National Meteorological Centres (NMCs) and transmitting them in the appropriate form on the Main Telecommunication Network (MTN), either directly or through the appropriate WMC/RTH in GTS until the transition from GTS to WIS 2.0 is advanced.

7.1.2. Regional Telecommunication Hubs

Regional Telecommunication Hubs (RTHs) shall keep their Message Switching Systems (MSS) up and continue to publish data Collecting the bulletins from their associated National Meteorological Centres (NMCs) and transmitting them in the appropriate form on the Main Telecommunication Network (MTN), either directly or through the appropriate WMC/RTH in GTS until all Members in their area of responsability migrate from GTS to WIS 2.0.

Regional Telecommunication Hubs (RTHs) that have implemented a WIS 2.0 node and all Members in their area of responsibility (AoR) have migrated to WIS 2.0, could decide to turn off their message switching system (MSS).

They shall liaise with their users and with centres in their AoR and point them to the WIS 2.0 Global Services.

7.1.3. National Meteorological Centres

National Meteorological Centres (NMCs) shall operate a WIS 2.0 node to share their data and discovery metadata in WIS 2.0.

NMCs that have implemented a WIS 2.0 node, and published all the data transmitted in the GTS on WIS 2.0 could, if they wish, turn off their GTS system MSS and stop transmitting data in the GTS. When NMCs decide to decommission and turn off their GTS system MSS and stop transmitting their data in GTS, they should include the GTS properties in the notification message as described in the WIS 2.0 to GTS gateway technical requirements.

Note: this notification message update will concern only data that are already published in GTS. New data will be published only on WIS 2.0.

7.2. Management of GTS Abbreviated Headings

7.2.1. GTS during the transition phase

- The GTS headers have been managed by volume C1 which has not been updated for several years
- The latest version of Volume C1 will be published on the WMO website (TBD)
- No new GTS header will be allocated during the transition phase
- The volume C1 will be decommissioned by December 2024

7.2.2. GTS Headings for ICAO (AFTN)

The management of GTS headings for the International Civil Aviation Organization (ICAO) will be done in collaboration with ICAO Whenever a new request is made to add a new CCCC, it will first be checked against the latest ICAO and WMO Secretariat CCCC list to ensure that it is not attributed to any centre, and then the WMO Secretariat will add it and publish it on the dedicated WMO website and inform ICAO about it

Chapter 8. Management of the WIS1 Catalogue

With the start of the pre-operational phase of WIS2, GISCs do no longer need to maintain and provide their own WIS1 Catalogue. In order to be able to decommission their copies of the Catalogue, GISCs SHALL liaise with their users and with centres in their AoR and point them to the WIS Portals of the WIS Catalogue Authorities, namely GISC Seoul and GISC Offenbach. GISC Seoul and GISC Offenbach will continue to provide WIS1 discovery metadata and the WIS1 Catalogue until the transition from GTS and WIS1 to WIS2 is advanced. ET-W2AT together with ET-OM propose to INFCOM the final date when the WIS1 Catalogue can be decomissioned in due time.

8.1. Management of Metadata

With the start of the pre-operational phase updates to WIS1 discovery metadata should be limited to important changes. Important changes are limited to: * updates to the points of contact * updates to the data formats / encoding * updates to the contents of the data Editoral changes or other updates to the discovery metadata for changes that do not affect data users will not be possible after the start of the pre-operational phase.

When GISCs decide to decomission and turn off their WIS1 catalogue and services they should arrange with GISC Offenbach and GISC Seoul how discovery metadata can be maintained. Also they should inform GISC Offenbach and GISC Seoul when their OAI interfaces become unavailable. GISCs may decide to continue to host their discovery metadata catalogue and their WIS1 services for the time being. In this case they should ensure the functioning of their respective OAI interfaces.

New discovery metadata should not be added to the WIS1 Catalogue any more. Instead new data should only be added in WCMP2 to the Global Discovery Catalogue for WIS2.

8.2. Global Discovery Catalogue

Users of the WIS1 discovery metadata catalogue should be informed by GISCs about the availability of WIS2 discovery metadata in the Global Discovery Catalogue. GISCs should provide guidance and asist their users in accessing and working with the WIS2 catalogue. With the start of the operational phase of WIS2, the Global Discovery Catalogues become the authoritative source for discovery metadata in WIS while the WIS1 catalogue will be terminated.

Chapter 9. References

9.1. Normative

- WMO: Manual on WIS (WMO No. 1060), Vol II. WIS 2.0
- WMO: WMO Core Metadata Profile version 2 (WCMP2) [1]
- WMO: WIS2 Topic Hierarchy (WTH) [2]
- WMO: WIS2 Notification Message (WNM) format [3]
- WMO: WIS2 Metric Hierarchy (WMH) [4]

9.2. Informative

- WMO: WMO Information System 2.0 Strategy (WMO No. 1213) [5]
- WMO: WMO Guidelines on Emerging Data Issues (WMO No. 1239) [6]

[1] https://wmo-im.github.io/wcmp2

[2] https://github.com/wmo-im/wis2-topic-hierarchy

[3] https://wmo-im.github.io/wis2-notification-message

[4] https://github.com/wmo-im/wis2-metric-hierarchy

[5] https://library.wmo.int/index.php?lvl=notice_display&id=20422

[6] https://library.wmo.int/index.php?lvl=notice_display&id=21644